Instructions for Use of CIPA Resolution Chart

1. Introduction

The CIPA Resolution Chart has been created in compliance with the specifications of ISO 12233:2017 (Photography -- Electronic still picture imaging -- Resolution and spatial frequency responses, hereafter referred to as "ISO12233:2017").

Comprehensive improvements have been made by the Camera & Imaging Products Association (CIPA) on the basis of the resolution chart given in ISO12233:2000 (hereafter referred to as the "Former ISO12233 Resolution Chart." See Fig. 1.), and this is referred to as the CIPA Resolution Chart in ISO 12233:2017.

The specifications of the "Former ISO12233 Resolution Chart" are also given in Annex I of ISO12233:2017.

If you are looking for the dealer of the CIPA Resolution Chart or the Former ISO12233 Resolution Chart, you will find it at the end of this document.

The ISO12233 standard is available for purchase at the Overseas Standard Section of (General Incorporated Association) Japanese Standards Association (http://www.jsa.or.jp/, Tel: +81-(0)3-3583-8000).

Concerning the preparation of this CIPA Resolution Chart, we have been given consent by Japan Electronics and Information Technology Industries Association (JEITA) (General Incorporated Association), which is the main body for deliberation of ISO12233 in Japan.

2. Relationship Between Standards and Items & Background

Item defined as a measurement item in CIPA DC-003

(1) Visual resolution measurement

Items defined as measurement items in ISO12233

(1) Visual resolution measurement

(2) e-SFR (Edge-based spatial frequency response measurement)

(3) s-SFR (Sine-based spatial frequency response measurement)

(limiting resolution measurement: This was given in ISO12233:2000 (first version) but was discontinued in the 2014 version.)

For details on the relationship and background of the above, refer to CIPA standard; Resolution Measurement Methods for Digital Cameras DC-003-2020.

3. Features of the CIPA Resolution Chart

This section describes the features of our CIPA Resolution Chart in comparison with the former ISO12233 Resolution Chart.

The CIPA Resolution Chart has the common features with the former ISO12233 Resolution Chart in that it is of a reflection type and the height of the effective area (inside the thick frame) of the chart is 200 mm.

3-1. Patterns for visual resolution measurement (J of Fig. 2)

These patterns are for measuring the visual resolution given in ISO12233:2017 and CIPA DC-003-2020.

a) Unified pattern for horizontal and vertical direction measurement

In response to the comment of some users suggesting that in the former ISO12233 Resolution Chart visual resolution patterns are not convenient because they are split into two, the CIPA Resolution Chart has one unified resolution pattern.

b) Unified pattern for 45-degree diagonal direction measurement

In the former ISO12233 Resolution Chart, the visual resolution pattern in 45-degree diagonal directions is divided into two, and further, these two are in different directions: the low-frequency pattern is sloping downward while the high-frequency pattern is sloping upward. Since the diagonal resolution of digital cameras may be different between upward and downward slopes, we have prepared both upward and downward patterns that are unified into one, similar to the case of horizontal and vertical directions.

c) Four directional patterns equidistantly positioned from the center

Generally, resolution of a lens (of camera) is the highest at the center and lowers toward the peripheral area. In the CIPA Resolution Chart, all four directional patterns are placed at equidistant positions from the center, allowing users to photograph and evaluate four types of visual resolution at once under the same conditions. We believe this layout is one of the most excellent features of our CIPA Resolution Chart.

d) Spatial frequency of the patterns

While the former ISO12233 Resolution Chart has the spatial frequency range of 100 to 2000 LW/PH, the CIPA Resolution Chart has the range of 200 to 2500 LW/PH in order to accommodate advanced digital camera technology.

e) Miscellaneous

The former ISO12233 Resolution Chart has five wedges in the low frequency pattern and nine wedges in the high frequency pattern. The CIPA Resolution Chart has five wedges each for both patterns for unification.

3-2. Patterns for SFR measurement (L of Fig. 2)

These patterns are for measuring e-SFR (edge - spatial frequency response) given in ISO12233:2017.

a) Unified pattern for measurement of horizontal and vertical directions

The former ISO12233 Resolution Chart uses different patterns for measurement of horizontal and vertical directions, but we have unified them in the CIPA Resolution Chart by adopting a square pattern.

b) Added pattern for 45-degree diagonal direction measurement

For SFR measurement in diagonal 45-degree directions, we have added a square pattern that can evaluate both upward and downward diagonals. We followed the former ISO12233 Resolution Chart in this respect.

c) Two patterns placed equidistantly from the center

Those two squares are positioned at equidistant positions from the center, allowing users to photograph and evaluate SFR in four directions at once under the same conditions.

(Remarks) Patterns for limiting resolution measurement (P of Fig. 2)

Measurement indices that use these patterns have been discontinued.

3-3. Extra features

a) Aspect ratio

The former ISO12233 Resolution Chart has an aspect ratio of 16:9. However, the CIPA Resolution Chart has the ratio of 3:2 because we decided most users would not require such a high length-to-width ratio.

b) Overall size

The A3 size (420mm x 297mm) has been adopted. We consider that this size would be convenient for delivery in generally available, standard size envelopes.

4. Accuracy of the chart

The range of tolerable errors for a resolution chart is described in ISO12233:2017 as follows:

a) The ratio of the maximum reflectance and the minimum reflectance (Rmax/Rmin) shall be defined by the ratio of reflectance of a large-area black pattern and white pattern, and shall hold the simultaneous inequality: 80 > Rmax/Rmin > 40. [Recommended] (5.2 of the Standards)

b) The positional tolerance shall be such that all test chart features are accurately located to within 0.2 mm, or ±0.1% of the active test chart height. [Required] (5.2 of the Standards)

c) The line width tolerance shall be within ±5%. [Required] (5.2 of the Standards)

d) The ratio of the maximum reflectance and the minimum reflectance (Rmax/Rmin) (for white part and black parts) at the finest features of the hyperbolic pattern K should be 18 or above. [Recommended] (Annex A of the Standards)

The CIPA Resolution Chart complies with all of these four requirements.

5. Conclusion

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Table 1: Description for the elements in Fig. 2

Element	Description
A	Frame define the active target area of the chart (use the inside area).
B	Arrows for checking horizontal alignment of the chart.
C	Indicates the center point of the chart, using an arrow wheel to facilitate focusing.
D	Arrows to define the aspect ratios of 1:1, 4:3 and 3:2.
J	Hyperbolic zone plates for visual resolution measurement.
K	Pattern for checking peripheral blurring; no scale provided because the pattern should be used only as a guideline.
L	Squares for e-SFR measurement. L1 for horizontal and vertical directions, and L2 for 45-degree diagonals.
P	Rectangular-wave multi-burst for measurement of limiting resolution. P1 is 200 to 1500 LW/PH, and P2 is 1400 to 2500 LW/PH.
Q	Concentric circles to help users position their own diagram when placing it on the chart.

(Prepared on March 8, 2022)

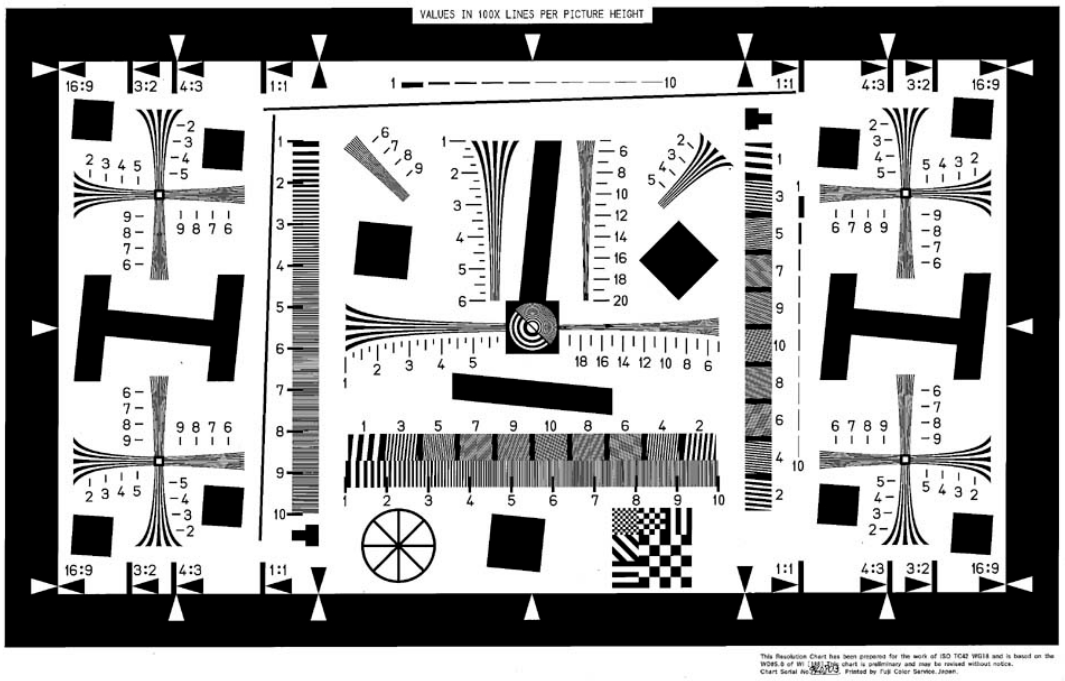


Fig. 1: Former ISO12233 Resolution Chart

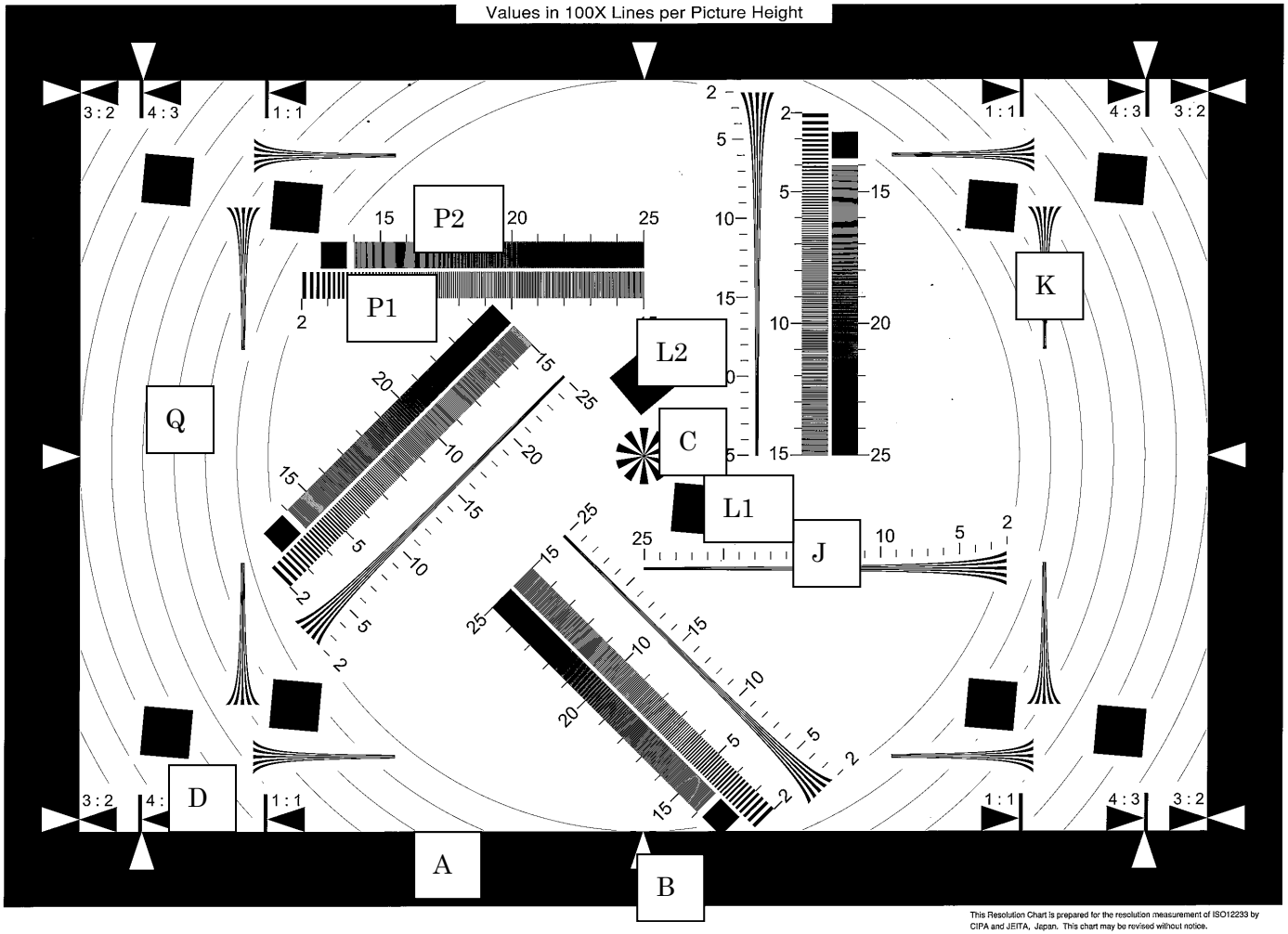


Fig. 2: CIPA Resolution Chart